Detecting Political Bias in News Articles Using Deep Learning:

From BiLSTM to Ordinal DistilRoBERTa

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Abstract

In this study, we set out to determine whether it is possible to identify the political leaning of a news article, categorized as *left*, *center*, or *right*, using only the article's text, without relying on any information about the source itself. To do this, we used the Article-Bias-Prediction dataset, which contains 37,554 labeled articles, and developed two supervised classifiers for comparison. Our starting point was a bidirectional LSTM (BiLSTM) with a learned embedding layer. This baseline model reached 61.4% accuracy on the three-way classification task, which is notably higher than the random baseline of 33.3%. After receiving feedback from reviewers, we pivoted our approach: an Ordinal DistilRoBERTa model with cumulative ordinal encoding. This method treats political labels as points along an ordered spectrum and penalizes predictions that are further from the true label more heavily. With this upgrade, the model's accuracy improved to 71.9%, and the balanced F1-scores for left, center, and right were 0.71, 0.72, and 0.72, respectively. To test how well these models generalize, we also evaluated them on 1,990 newly collected articles from six major news outlets (BBC, ABC, CBC, FOX, Al Jazeera, and NPR). Interestingly, the BiLSTM struggled with FOX News articles, often misclassifying them as left-leaning. The RoBERTa model addressed this issue, increasing the proportion of FOX articles correctly identified as right-leaning from 11.6% to 74.9%. Overall, our findings indicate that political leaning can be learned from article text alone, but achieving strong generalization depends on both the diversity of training sources and the depth of the model's representations.

Introduction

"A 2019 survey found that more than 80% of Americans felt there was "a great deal" or "a fair amount" of political bias in the news" (Knight Foundation, 2020). Readers already notice this distinction between news articles today but the implications of this extend beyond individual readers. Biased articles frequently make their way into the training data of large language models, shaping what AI systems learn and how they respond before the model is ever deployed. At the same time, biased reporting continues to influence the public directly given how central and accessible media is.

The phrase "political bias in media" comes up often, but it is not always defined clearly. Most tools that aim to measure bias, such as the Ad Fontes Media Bias Chart and AllSides, tend to classify entire news outlets as left, center, or right. The problem with this approach is that a right-leaning outlet might publish balanced or centrist reporting on some topics, while a centrist outlet might sometimes run opinion-heavy pieces. When we apply these broad source-level labels to individual articles we risk introducing systematic errors.

In this project, we are interested in a more detailed question: is it possible to identify an article's political leaning solely from its text, without any information about the source? This idea is based on the Distributional Hypothesis, which states that *words used in similar contexts tend to have similar meanings*. If that is true, then it stands to reason that the language and structure of an article might reveal its ideological slant.

What sets this project apart are three main things. First, the analysis is conducted at the article level rather than the source level. Second, we compared two very different modeling approaches: a BiLSTM built from the ground up, and a fine-tuned DistilRoBERTa transformer. Third, we used ordinal encoding to capture the natural order of the left-center-right spectrum. This way confusing left with right is treated as a bigger mistake than confusing left with center. To test how well these models perform, we also validated them on real-world articles not included in the training data, thereby demonstrating their generalization.

Related Work

Source-level bias classification. Baly et al. (2018) laid important groundwork for classifying news sources by both factuality and political leaning, drawing on information from Wikipedia pages and Twitter metadata. While their approach works well at the source level, it does not address the challenge of predicting ideology at the article level. The model needs to pick up on ideological cues without relying on the outlet that published the piece.

The Article-Bias-Prediction dataset. Baly et al. (2020) introduced the dataset we use in this project, providing article-level political ideology labels (left, center, right) derived from media bias assessments. Their supervised classification approach and dataset construction methodology directly informed our experimental setup.

Transformer-based approaches. Loiya et al. (2026) introduce ChunkyBERT, a BERT-based chunking classifier designed for detecting political bias in longer texts. While they achieve impressive, state-of-the-art results, their method is much more computationally intensive than what we are aiming for. Their findings encourage us to consider transformer architectures as a possible next step, but we are mindful of the resource trade-offs.

Word embeddings and ideology. Rheault and Cochrane (2019) show that word embeddings trained on parliamentary texts can uncover underlying ideological patterns, which supports the Distributional Hypothesis in the context of political language. This theoretical backing is one of the reasons we decided to start with an embedding-based approach in our own work.

Ordinal classification. Ordinal encoding schemes, which account for the order of labels, have been used in NLP tasks such as essay scoring and sentiment analysis. In our project, we use a cumulative encoding approach (left as (1, 0, 0), center as (1, 1, 0), and right as (1, 1, 1)) to better capture the idea of ideological distance in our loss function.

Hybrid architectures. Tan et al. (2022) combine RoBERTa with an LSTM in a hybrid model for sentiment analysis, achieving F1-scores between 90 and 93 percent across several datasets. Their results suggest that using transformer-based contextual embeddings together with an LSTM to model sequence information can outperform either method alone. This is a promising direction we may want to explore in future work on this task.

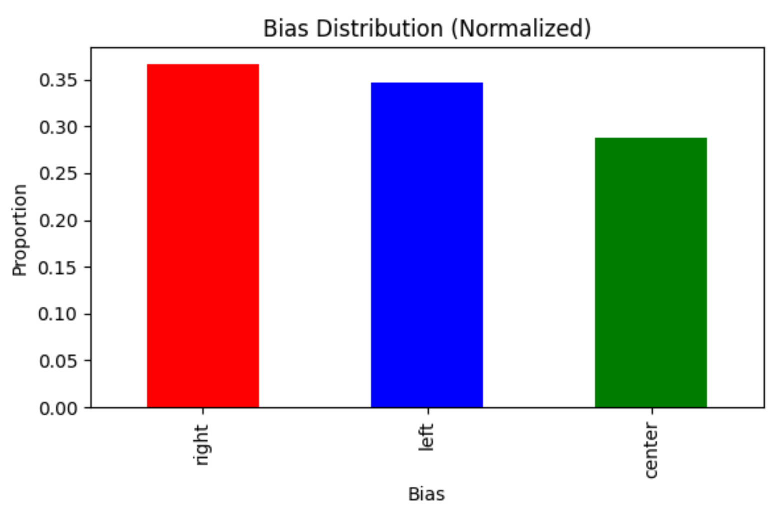
Data

3.1 Training Dataset

Table 1. Article-Bias-Prediction dataset statistics.

Split	Left	Center	Right	Total
Train	9,750	7,988	10,240	27,978
Validation	2,438	1,998	2,560	6,996
Test	402	299	599	1,300
Total	**12,590**	**10,285**	**13,399**	**36,274**

The dataset includes 37,554 articles, each described by 12 columns, including article content, source name, and bias label. Notably, there are no missing values, which helps ensure the reliability of any analysis. On average, articles are 1,084 words long, though the range is quite broad, from just 26 to 9,703 words. Importantly, the word-count distributions are similar across all three bias classes, so article length is unlikely to skew the results. The classes themselves are fairly balanced, with 36.6% labeled as right, 34.6% as left, and 28.8% as center.



3.2 Real-World Evaluation Corpus

To assess how well the model generalizes beyond the training data we gathered 1,990 articles from six news sources on May 25, 2026, via web scraping. These sources, BBC (727 articles), ABC (417), CBC (287), FOX (259), Al Jazeera (220), and NPR (80), span a range of perspectives according to the Media Bias Chart, and none were included in the training set. Since we do not have ground-truth labels for these articles, this evaluation serves as a qualitative check of generalization rather than a strict accuracy test.

Approach

4.1 Experiment 1: Bidirectional LSTM Baseline

Preprocessing. To begin, all article text is converted to lowercase. We also removed URLs, email addresses, special characters, and digits using regular expressions, and then collapsed any extra whitespace. This preprocessing step helps ensure that the model focuses on the *core textual content* without distractions from formatting or irrelevant tokens.

Tokenization. Next, we fit a Keras Tokenizer on the training corpus, keeping only the 5,000 most frequent tokens to limit the vocabulary size. Each article is then represented as a sequence of integers, which are either padded or truncated to a maximum length of 500 tokens. We used post-padding and post-truncation so that all sequences are the same length, which is necessary for batch processing in neural networks.

Architecture. The model is implemented in TensorFlow/Keras with the following layer stack:

1. Embedding layer: vocabulary size 5,000, output dimension 128, learned from scratch
2. Dropout (rate = 0.5)
3. Bidirectional LSTM (64 units, return_sequences=True)
4. Dropout (rate = 0.5)
5. Bidirectional LSTM (64 units)
6. Dense (128 units, ReLU) → Dropout (0.5)
7. Dense (64 units, ReLU) → Dropout (0.5)
8. Output Dense (3 units, softmax)

Training. For training, we used the Adam optimizer with a learning rate of 0.001 and sparse categorical cross-entropy as the loss function. The data is split into 80% training and 20% test sets, stratified by class to maintain balance. It trains for up to 15 epochs, but uses early stopping based on validation loss with a patience of 3, restoring the best weights if performance plateaus.

Evaluation. To evaluate the model, we examined overall accuracy, per-class precision, recall, F1 scores, and the confusion matrix to gain a detailed sense of performance across categories.

4.2 Experiment 2: Ordinal DistilRoBERTa

This experiment focuses on two main points raised by reviewers. First, LSTM models learn their own embeddings from scratch, which can be a disadvantage compared to transformer models that use pretrained contextual representations. Second, the standard softmax classification head does not account for the fact that some label errors are more severe than others, since it treats all mistakes as equally costly and ignores the natural order of the political spectrum.

Ordinal encoding. To address the ordinal nature of the labels, we encoded them as cumulative binary vectors: left is (1,0,0), center is (1,1,0), and right is (1,1,1). During inference, we applied a sigmoid to each logit, applied a threshold at 0.5, and counted the cumulative number of active bits to recover the predicted class. This approach means that making a mistake between left and right is penalized more heavily than confusing left with center or center with right, since opposite-pole errors require more incorrect ordinal decisions.

Base model. For the model, we used distilroberta-base from the HuggingFace transformers library. This is a lighter version of the RoBERTa family, with 82.1 million parameters. We used byte-level Byte-Pair Encoding for subword tokenization and set the maximum sequence length to 128 tokens. The encoder is fully fine-tuned, rather than frozen, so it can adapt to the specifics of the task.

Architecture. The [CLS] token embedding from the final transformer layer is sent through a dropout layer with a rate of 0.1. After this, it is fed into a linear classifier that produces three ordinal output logits. For the loss function, we used BCEWithLogitsLoss, applying it separately to each of the three ordinal bits.

Training. We implemented the training pipeline using PyTorch and the HuggingFace Trainer API. For optimization, we used AdamW with a learning rate of 2e-5 and a weight decay of 0.01. To manage memory, we used gradient accumulation with an effective batch size of 8 and trained for 3 epochs. The best model checkpoint is chosen based on the lowest validation ordinal MAE. Training is done on an NVIDIA GPU with CUDA 13.0.

Additional evaluation metrics. In addition to accuracy and per-class F1, we report: (1) Ordinal MAE: mean absolute error on class IDs (0=left, 1=center, 2=right), where adjacent errors score 1 and opposite-pole errors score 2; (2) Ordinal MSE: mean squared error on class IDs; (3) Severe error rate: proportion of predictions where |y_true − y_pred| = 2.

Experiment and Results

5.1 BiLSTM Baseline

The model converged in just 7 epochs, with early stopping restoring the best weights from epoch 4. At that point, validation accuracy reached 61.4% and validation loss was 0.861. Training accuracy at best epoch was 69.8%, which suggests the model was able to learn meaningful patterns from the data without falling into severe overfitting. When evaluated on the separate test set of 7,511 articles, the BiLSTM achieved 61.4% accuracy. For context, this is a substantial improvement over the random baseline of 33.3%.

Table 2. BiLSTM per-class performance on the test set.

Class	Precision	Recall	F1	Support
Left	0.62	0.61	0.61	2,601
Center	0.65	0.52	0.58	2,163
Right	0.59	0.70	0.64	2,747
Macro avg	**0.62**	**0.61**	**0.61**	**7,511**

Notably, the model struggled most with center articles, which had the lowest F1 score at 0.58. This outcome is not surprising, since centrist writing often shares vocabulary with both left and right-leaning articles, making it harder for the model to distinguish. The confusion matrix supports this, showing that center articles were frequently misclassified as either left or right.

5.2 Ordinal DistilRoBERTa

Training proceeded for 3 epochs with consistent improvement in validation metrics.

Table 3. Ordinal DistilRoBERTa training progression.

Epoch	Train Loss	Val Loss	Accuracy	Ord. MAE	Severe Err.
1	0.391	0.274	0.741	0.326	0.068
2	0.405	0.230	0.780	0.288	0.068
3	**0.369**	**0.255**	**0.789**	**0.276**	**0.065**

On the test set of 1,300 articles, the model achieved 71.9% accuracy. Per-class performance and distance-aware metrics are reported below.

Table 4. Ordinal DistilRoBERTa per-class performance on the test set.

Class	Precision	Recall	F1	Support
Left	0.76	0.67	0.71	402
Center	0.58	0.96	0.72	299
Right	0.85	0.63	0.72	599
Macro avg	**0.73**	**0.76**	**0.72**	**1,300**

Distance-aware metrics: Ordinal MAE: 0.390 (on a 0–2 scale) · Ordinal MSE: 0.608 · Severe error rate: 10.9% · Ordinal error distribution: 935 exact, 223 adjacent-class, 142 opposite-pole.

5.3 Model Comparison

Table 5. Summary comparison of both models on the test set.

Model	Accuracy	Left F1	Center F1	Right F1	Ord. MAE	Severe Err.
BiLSTM baseline	0.614	0.61	0.58	0.64	N/A	N/A
Ordinal DistilRoBERTa	0.719	0.71	0.72	0.72	0.39	10.9%
Improvement	**+10.5pp**	**+0.10**	**+0.14**	**+0.08**	**—**	**—**

The most significant improvement came with center articles (+0.14 F1). Since these were the most challenging for the BiLSTM, this gain is particularly meaningful. The RoBERTa model achieved a center recall of 0.96, indicating it identified nearly all centrist articles. However, the precision for center remained at 0.58, indicating that the model tended to over-predict the center label when it was unsure.

5.4 Real-World Generalization

When the BiLSTM was applied to a new set of 1,990 recently collected articles, it predicted that about half (49.9%) were center, 31.9% were left, and 18.2% were right. The average confidence in these predictions was 64.5%, but it is worth noting that roughly 28% of the articles received a confidence score below 0.5, suggesting some uncertainty in the model's classifications.

One of the most notable results involves FOX News. The BiLSTM model classified 51.4% of FOX articles as left and only 11.6% as right, which is counterintuitive given FOX's reputation. In contrast, the Ordinal DistilRoBERTa model almost completely reversed these numbers, predicting 74.9% of FOX articles as right and just 8.5% as left. For NPR, the proportion of articles labeled as center increased from 31.3% with the BiLSTM to 64.7% with RoBERTa, which aligns much more closely with NPR's widely recognized centrist stance.

Table 6. Real-world source prediction comparison between models.

Source Metric	BiLSTM Baseline	Ordinal DistilRoBERTa
FOX right share	11.6%	74.9%
FOX left share	51.4%	8.5%
NPR center share	**31.3%**	**64.7%**

Discussion

6.1 Findings and Insights

The main takeaway here is that it is possible to detect political leaning simply by analyzing an article's text. Both models performed much better than we would expect by chance, which is about 33.3%. The improved performance of the RoBERTa model shows that as the model's ability to represent language becomes more sophisticated, it can pick up on these signals with even greater accuracy.

One of the most interesting results is how the BiLSTM model misclassified FOX articles. This suggests that the BiLSTM was picking up on specific language patterns tied to particular sources, rather than learning a broader understanding of political ideology. The model relied on surface-level similarities in vocabulary and topics that happened to overlap with those of left-leaning sources. In contrast, the RoBERTa model, which uses pretrained contextual embeddings, was able to develop a more general understanding and handled the FOX articles much better.

Beyond accuracy, ordinal metrics provide a more detailed picture of model performance. For example, the ordinal mean absolute error (MAE) is 0.39 on a scale from 0 to 2, which means that, on average, the model's predictions are less than half a step away from the true ideological label. Importantly, only about 11% of predictions are completely off, landing on the opposite end of the spectrum. Most of the time, when the model makes a mistake, it predicts a neighboring class rather than jumping across the entire spectrum. This is exactly what the ordinal encoding was meant to encourage.

One area where the model still struggles is with predicting centrist articles. While the recall for the center class is high at 0.96, the precision is much lower at 0.58. This means the model often predicts 'center' when it is unsure, which makes sense because centrist articles tend to use language that overlaps with both left- and right-leaning articles. As a result, it is harder for the model to confidently identify truly centrist content.

6.2 Limitations and Future Work

- **Training distribution.** The Article-Bias-Prediction dataset is built from a fairly narrow selection of sources. Although upgrading to RoBERTa led to a noticeable improvement in generalization, it is important to note that the model has not actually seen data from outlets like FOX or NPR. Expanding the dataset to include a wider range of ideologically diverse sources would likely help the model generalize even better.
- **Sequence truncation.** DistilRoBERTa is limited to processing just 128 tokens per article, which means that for the average article length of 1,084 words, the model is only exposed to roughly the first 100 words. As a result, much of the article's content is left out of the analysis. Implementing a chunking strategy or using hierarchical encoding could allow the model to take more of the article into account, potentially leading to more accurate predictions.
- **Label granularity.** Using only three classes to categorize articles is a fairly coarse approach. By moving toward a more fine-grained ideological scale, such as the two-dimensional reliability and bias axes used by Ad Fontes, it may be possible to capture subtleties and nuances that a simple three-way split tends to overlook.
- **Center over-prediction.** The precision for the center class remains relatively low at 0.58. Adjusting thresholds for each class individually or calibrating the model's output layer could help address this issue and improve overall performance.
- **Future directions.** There are several promising directions for future work. These include fully fine-tuning RoBERTa-large, experimenting with a hybrid RoBERTa-LSTM architecture as described by Tan et al. (2022), applying the model to social media texts such as tweets or Reddit posts, and incorporating confidence calibration to make deployment safer and more reliable.

Ethical Considerations

Reductive labeling. The common practice of dividing political ideology into left, center, and right categories tends to oversimplify a complex, multidimensional landscape. While this framework is often used to make sense of political differences, it can reinforce the very polarization it aims to describe. Many political perspectives, such as libertarianism, green politics, or religious conservatism, do not fit neatly into this left-right spectrum, which means they are often misrepresented or overlooked by such classifications.

Training data bias. The labels used in the Article-Bias-Prediction dataset come from external media bias assessments, but it is important to recognize that these assessments are not entirely objective. The organizations responsible for these judgments bring their own cultural, geographic, and institutional perspectives to the table, which can influence how sources are categorized. When we apply these source-level labels to individual articles, we introduce yet another layer of uncertainty, making it even more challenging to draw clear conclusions about bias.

Misclassification risk. The misclassification of FOX articles by our BiLSTM model highlights the potential risks of deploying such systems. If a tool incorrectly and confidently labels FOX articles as 'left,' it could easily mislead users, undermine trust in automated analysis, or even be used as ammunition in ongoing debates about media bias. While the ordinal RoBERTa model addresses this particular issue, it is important to keep in mind that similar problems could arise with other sources that fall outside the training distribution.

Responsible deployment. In practice, any real-world use of this classifier should be approached with caution. It is essential to set confidence thresholds to filter out unreliable predictions, provide clear documentation of the training data and its origins, and be upfront about the model's limitations. Additionally, outputs should be presented as probabilistic predictions rather than definitive statements, so users understand that these results are not absolute.

Positionality. This project was conducted within a US academic setting, and the dataset we used mostly comes from US English-language news sources. Because of that, the way we frame things as left, center, or right is based on US political conversations, which might not line up with how politics are structured in other countries.

Conclusions

Our findings show that it is possible to detect political leaning simply by analyzing an article's text. Both the BiLSTM baseline (61.4% accuracy) and the Ordinal DistilRoBERTa model (71.9% accuracy) performed well above random chance. After receiving feedback, we moved from BiLSTM to RoBERTa, which led to higher F1-scores for all three classes, especially for center articles, where the score increased by 0.14. By adding ordinal encoding, we also saw a more balanced error distribution, with only 10.9% of predictions falling into the most severe category of opposite-pole errors.

One of the most interesting takeaways from this study is the story behind the FOX misclassification. This example highlights the difference between models that capture specific language patterns from certain sources, such as BiLSTM, and those that learn broader ideological cues, such as RoBERTa. It also shows the real risks involved when bias classifiers are trained on a narrow set of sources. By changing the model, we were able to address this issue directly, offering clear evidence that expanding the training data and deepening the model’s understanding are key to improving performance in this area.

This work provides a reproducible baseline for article-level political bias classification, along with a documented upgrade path from simple sequence models to ordinal transformer classifiers.

References

Baly, R., Karadzhov, G., and Alexandrov, D. (2018). Predicting factuality of reporting and bias of news media sources. In Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing (EMNLP). https://aclanthology.org/D18-1389/

Baly, R., Da San Martino, G., Glass, J., and Nakov, P. (2020). We can detect your bias: Predicting the political ideology of news articles. In Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP). https://aclanthology.org/2020.emnlp-main.404

Knight Foundation. (2020, November 9). *American views in 2020: Trust, media and democracy*. https://knightfoundation.org/wp-content/uploads/2020/08/American-Views-2020-Trust-Media-and-Democracy.pdf

Loiya, D., Kulal, S.S., Reddy, M.S.M., et al. (2026). ChunkyBERT: A novel technique for multiclass political bias detection in news media. Scientific Reports. https://doi.org/10.1038/s41598-026-46646-z

Rheault, L., and Cochrane, C. (2019). Word embeddings for the analysis of ideological placement in parliamentary corpora. Political Analysis, 28(1), 112–133.

Tan, K.L., Lee, C.P., Anbananthen, K.S.M., and Lim, K.M. (2022). RoBERTa-LSTM: A hybrid model for sentiment analysis with transformer and recurrent neural network. IEEE Access, 10, 21517–21525. https://doi.org/10.1109/ACCESS.2022.3152828

Appendix - Training Dataset

News Source	Count
CNN (Web News)	2905
Politico	2493
NPR Online News	2012
USA TODAY	1791
New York Times - News	1417
The Hill	1377
Christian Science Monitor	1300
Reuters	927
BBC News	886
The Guardian	644
Associated Press	391

ABC News	341
Wall Street Journal - News	335
Bloomberg	178
The Atlantic	174
CBS News	164
Al Jazeera	145
Washington Post	109
CNBC	108
Axios	76
Time Magazine	75
ProPublica	72
NBC News (Online)	38
The Economist	28
New York Times (Online News)	22
Foreign Policy	20
PBS NewsHour	14